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RESEARCH ARTICLE

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Nitrogen-doped carbon dots as a fluorescent probe for the highly sensitive detection of bilirubin and cell imaging

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Abstract

Nitrogen-doped carbon dots (NCDs) with bright blue fluorescence were constructed by a hydrothermal method using sucrose and L-proline as raw materials. The NCDs were characterized by transmitted electron microscopy, X-ray diffraction, Fourier-transform infrared spectrometry, X-ray photoelectron spectroscopy, and ultraviolet-visible absorption and fluorescence spectroscopy to investigate the morphology, elemental composition, and optical properties. The NCDs had good water solubility, high dispersibility with an average diameter of only 1.7 nm, and satisfactory optical properties with a fluorescence quantum yield of 23.4%. The NCDs were employed for the detection of bilirubin. A good linear response of the NCDs in the range 0.35–9.78 μM was obtained for bilirubin with a detection limit of 33 nM. The NCDs were also applied to the analysis of real samples, serum and urine, with a recovery of 95.34% to 104.66%. The low cytotoxicity and good biocompatibility of the NCDs were indicated by an MTT assay and cell imaging of HeLa cells. Compared with other detection systems, using NCDs for bilirubin detection was a facile and efficient method with good selectivity and sensitivity.

KEYWORDS
bilirubin, biocompatibility, fluorescent probe, nitrogen-doped carbon dots, static quenching

1 | INTRODUCTION

Carbon dots (CDs) are a new member of the carbon nanomaterial family that have a diameter < 10 nm. In recent years, CDs have received much attention because of their cost-effective preparation, large specific surface area, optical properties, low toxicity, and biocompatibility.[1–4] CDs have been widely used in bioimaging,[5] biosensors,[6] photoelectric catalysis,[7] nanomedicines,[8] drug transport,[9] and environmental detection.[10] Heteroatom (N, P, S, Fe, and Ag) doping of CDs can increase the quantum yields and provide CDs with more stable optical properties.[11–13] Thus, the green, facile and highly efficient synthesis of doped CDs for the detection of biomolecules is desirable.

Bilirubin is a waste product of metabolized hemoglobin in the blood. Although bilirubin can be used as an endogenous antioxidant to prevent the oxidation of phospholipids and linoleic acid, it can also cause damage to the brain and nervous system.[14,15] When the total bilirubin content is higher than 34 mol/L, hearing loss, brain damage, jaundice, cirrhosis of the liver, and other symptoms can occur and serious conditions can be life-threatening.[16–18] Therefore, a method for the simple, accurate, and inexpensive determination of bilirubin is becoming increasingly important in medicine and pharmacology. To date, many chemical techniques have been proposed to determine the bilirubin content, including high-performance liquid chromatography,[19] capillary electrophoresis,[20] chemiluminescence,[21] Raman spectroscopy,[22] electrochemical methods,[23] and ultraviolet (UV) spectrophotography.[24] However, the earlier-mentioned methods suffer from certain drawbacks, such as the high cost and time-consuming pre-processing requirements, poor sensitivity and selectivity, and long response times. Fluorescence

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analysis using CDs is a promising platform for chemical, biological, and environmental detection with high sensitivity and selectivity.[25–27] However, the development of suitable CDs for bilirubin detection has not been explored in depth. Anjana et al.[28] have reported an effective fluorescence probe for the detection of bilirubin using sulfur (S)- and nitrogen (N)-doped CDs, which were synthesized using citric acid and L-cysteine by a microwave-assisted method. Recently, Ottoor and coworkers[29] have synthesized CDs from sunflower seeds, which were used to determine bilirubin content with a limit of detection of 2.03 μM. These synthetic doped CDs showed a moderate quantum yield and a high detection limit. Hence, developing doped CDs as an effective fluorescence probe for the detection of bilirubin is highly feasible.

In this work, N-doped carbon dots (NCDs) with a high quantum yield of 23.4% were successfully synthesized from sucrose and L-proline by a hydrothermal method and characterized by different techniques. The NCDs showed good sensitivity and selectivity for bilirubin detection with a detection limit of 33 nM. The NCDs were used for the detection of bilirubin in serum and urine. In addition, the NCDs showed low toxicity and good biocompatibility, which were determined by an MTT assay and the cell imaging of HeLa cells.

2 | EXPERIMENTAL

2.1 | Materials and instruments

All reagents were analytically pure and purchased from Aladdin. Deionized water was used in the experiment. The morphology was determined on a JEOL JEM-2100F transmission electron microscope (Tokyo, Japan). Bruker Tensor II Fourier infrared spectrometer (Bremen, Germany) was used to analyze the functional group. X-ray photoelectron spectroscopy (XPS) was carried out with a Kratos AXIS ULTRA DLD spectrometer (Tokyo, Japan). The absorption spectrum was recorded using aHitachi UV-2910 UV spectrophotometer (Tokyo, Japan). Fluorescence experiments were measured with a Hitachi F-4500 fluorescent spectrophotometer (Tokyo, Japan). Digital pH meter SJ-3F (Leigu, Shanghai, China) was used to adjust the pH of the phosphate-buffered saline (PBS). The fluorescent images of HeLa cells were performed on an Olympus FV1000-MPE multiphoton laser scanning microscope (Tokyo, Japan). The dry samples were obtained in an FD-1D-50 freeze vacuum dryer (Boyikang, Beijing, China).

2.2 | Synthesis of NCDs

For the preparation of the NCDs, 0.5940 g of sucrose and 0.5855 g of L-proline were mixed in a 50-mL flask and dissolved in 30.0 mL of water with vigerous stirring. The homogeneous mixture was then transferred into a 50-mL autoclave and kept at 200°C for 8 h in an oven. After the autoclave was naturally cooled to room temperature, the NCD solution was filtered with a 0.22-μm microporous membrane. Then, dialysis (molecular weight cut-off of 500 D) against distilled water for 48 h was used to remove any residual species. Finally, solid residues were obtained by freeze-drying the purified yellow solution under a vacuum.

2.3 | Detection of fluorescence quantum yield

The fluorescent quantum yield of the NCD solution was measured by reference to a quinine sulfate solution (0.1 M) using an UV spectrophotometer to ensure that the absorbance value of the quinine sulfate and the NCD solution at 360 nm was < 0.05. The quantum yields of the NCDs and quinine sulfate were calculated using the equation ($QY_R = 0.54$, $\eta_R = 1.33$).

$$QY = QY_R \times \frac{I}{I_R} \times \frac{A_R}{A} \times \frac{\eta^2}{{\eta_R}^2}$$

where QY and QY_R represent the fluorescent quantum yields of the sample and the reference solution, respectively; I and I_R represent the integral areas of the fluorescence spectrum of the sample and the reference solution, respectively; A and A_R are the UV-visible absorbent values of the sample and the reference solution, respectively; and η and η_R indicate the refractive index of the sample and the reference solution, respectively (the refractive index of the sulfuric acid solution and water in this experiment were the same).

2.4 | Detection of bilirubin and interference experiments

For the fluorometric detection of bilirubin, 80 μL of NCD solution was poured into a 10-mm quartz cell, then different volumes of bilirubin solution (6.98×10^{-5} mol/L) were added and the final volume of the mixture was made up to 2.0 mL with PBS (10 mM, pH 7.80). The solution was gently shaken and reacted for 10 min at room temperature. The fluorescence spectrum was measured at an excitation wavelength of 370 nm by setting the slit width to 5 nm. The responses were monitored from 390 to 720 nm. For selectivity experiments, the common metal ions (Mg^{2+}, Na^{+}, Ca^{2+}, Zn^{2+}, Fe^{3+}, and K^{+}) and biomolecules (glucose, fructose, glutathione, dopamine, L-phenylalanine, L-glycine, vitamin C, creatine, uric acid, lactose, thiourea, urea, sucrose, and starch) replaced the bilirubin in the system and the experiments were performed using the same conditions.[30]

2.5 | Detection of bilirubin in biological samples

Serum samples were collected from clinical laboratories, and urine samples were collected from a healthy volunteer in our laboratory, neither type of sample contained any bilirubin. Both types of samples were first filtered with a 0.22-μm membrane, then centrifuged for 30 min (8000 rpm). Then, the supernatant was separated to fabricate

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spiked samples and stored at 4°C. The supernatant was diluted 100 times with PBS buffer solution (10 mmol/L, pH 7.80) before use.

2.6 | Cytotoxicity test for NCDs

HeLa cells were incubated under standard conditions (37°C and 5% carbon dioxide). When the cells covered the wall of the culture flask, the cells were digested and shifted to a 96-well plate. The cell density of each well reached 5×10^3 cells/mL and the number of cells was recorded by a cell count plate. Different concentrations of NCDs (0, 5, 10, 70, 80, 90, 700, 800, 900, and 1,000 μg/mL) were introduced into the cells and the cells were incubated for 24 h. Then, 20 μL of MTT (0.5 mg/mL) was added to the cells and the cells were incubated for 4 h. Finally, the MTT was replaced with 150 μL of dimethyl sulfoxide (DMSO) in each well. After mixing for 15 min at room temperature, the absorbance was determined using a microplate reader.[31]

2.7 | Cell imaging

HeLa cells were suspended in a 35-mm Petri dish with 10-mm holes at the bottom and cultured for 24 h. After washing with PBS buffer, the HeLa cells were incubated with 0.4 mg/mL NCDs at 37°C for 1.5 h (control group). In the experimental group, 20 μM bilirubin was added to the cells and the cells were incubated for 1.5 h. After removing the supernatant, the cells were washed with PBS and then fixed with 4% *p*-formaldehyde for 10 min. Finally, the fluorescence images were recorded using a confocal laser scanning microscope.

3 | RESULTS AND DISCUSSION

3.1 | Preparation of NCDs

The NCDs were prepared using sucrose and L-proline as raw materials by a hydrothermal method. The reaction temperature, reaction time, and the molar ratios of the raw materials were optimized in terms of the fluorescence properties (Supporting Information Figure S1). The maximum fluorescence emission for the NCDs was obtained by the reaction of a 1:1 molar ratio of sucrose and L-proline at 200°C for 8 h measured at 437 nm.

3.2 | Characterization of the NCDs

The prepared NCDs were well dispersed in water and had bright blue fluorescence under UV light at 365 nm. The transmission electron microscopy (TEM) images revealed that the NCDs were spherical with an average particle size of 1.7 nm [Figure 1 (inset)], which indicated a good dispersion.

Figure 2a shows the X-ray diffraction (XRD) pattern of the NCDs, where a wide peak was observed at $2\theta = 23.48°$, which was

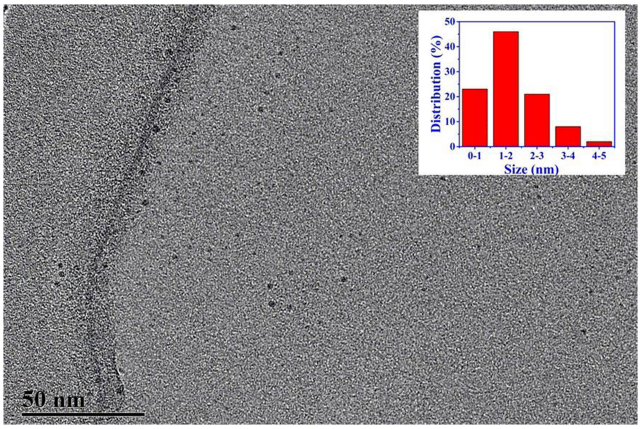

FIGURE 1 TEM images of the NCDs (inset: the size distribution of the NCDs)

consistent with the crystalline graphite of the CDs.[32] The Fourier-transform infrared (FTIR) spectrum shown in Figure 2b shows the surface functional groups of the NCDs. The wide absorption peak at 3417 cm^{-1} corresponds to the stretching vibration peak of the O–H/N–H bond.[33] The stretching vibration peak at 2869 cm^{-1} is attributed to the C–H bond of an alkyl group.[34] The stretching vibration peak of the C=O bond appears at 1619 cm^{-1} and the in-plane bending vibration peaks of C–H and C–N appear at 1384 and 1,450 cm^{-1}, respectively.[35,36]

The elemental compositions of the NCDs determined by XPS are shown in Figure 3. Figure 3a shows the absorption peaks at 532.5, 399.3, and 284.7 eV, corresponding to the binding energy of O1s, N1s, and C1s respectively, indicating that the NCDs contain C, N, and O atoms.[37] The C1s spectrum in Figure 3b shows peaks at 284.6, 285.7, 287.4, and 288.4 eV, which indicates the presence of C=C, C–N, C–O, and C=O groups, respectively.[38] As can be seen from Figure 3c, the N1s spectrum was consistent with N–H/C–N (399.3 eV) and pyridine nitrogen (400.1 eV).[39] The O1s spectra in Figure 3d indicated the presence of C–OH/C–O–C (531.3 eV), H–O–H (532.2 eV), and C=O (536.2 eV).[40]

The UV-visible absorption and fluorescent excitation/emission spectra of the NCDs are shown in Figure 4. The maximum UV-visible absorption peak of the NCDs was at approximately 330 nm, which was caused by the n-π* transition of the C=O group.[41] When the excitation wavelength was 370 nm, the maximum emission wavelength was 437 nm (Figure 4a). The solution of the NCDs appeared light yellow under a daylight lamp and emitted bright blue fluorescence under UV light (inset, $\lambda_{ex} = 365$ nm), which was consistent with the emission spectrum of the NCDs. In addition, we investigated the excitation-dependent behavior at different excitation wavelengths. As shown in Figure 4b, when the excitation wavelength changed from 300 to 370 nm, the fluorescence intensity increased continuously; however, the fluorescence intensity decreased with an increase in excitation wavelengths from 370 to 430 nm. This result indicated that the emission wavelength depended on the excitation wavelength. The blueshift of the peak position may be related to the

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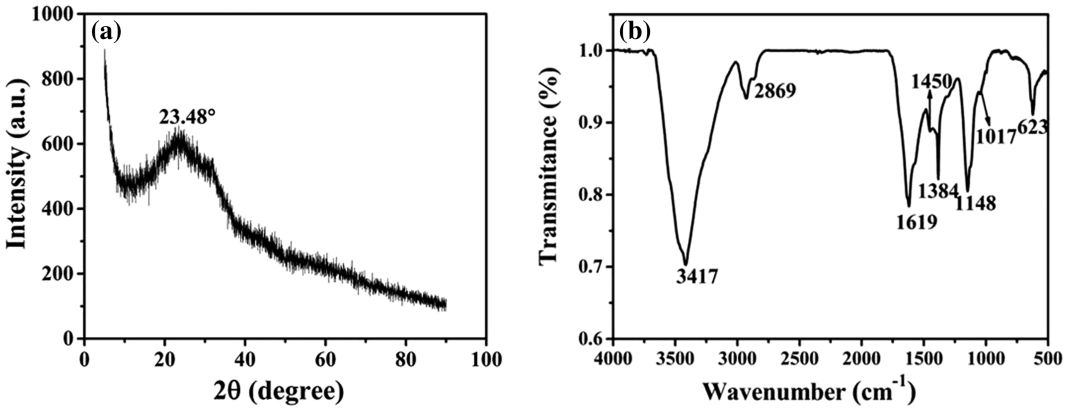

FIGURE 2 (a) XRD spectrum of NCDs; (b) FTIR spectrum of NCDs

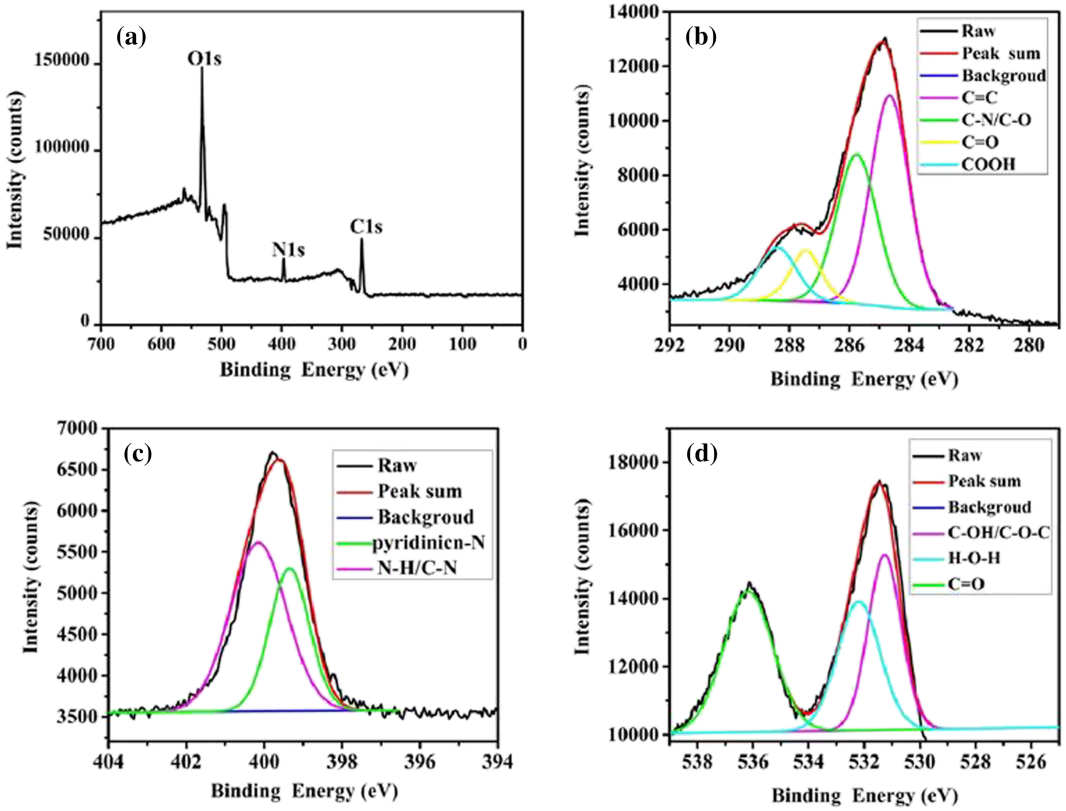

FIGURE 3 (a) XPS full spectrum of the NCDs; (b) C1s; (c) N1s; (d) O1s

particle sizes and different emission sites on the surface of the NCDs.[42] The maximum emission peak (437 nm) was observed when the excitation wavelength was 370 nm. An excitation wavelength of 370 nm was selected for subsequent experiments. The fluorescence quantum yield of the NCDs was calculated to be 23.4%, using quinine sulfate as a standard, which was higher than the reported fluorescence quantum yield from natural sources.[29] UV radiation and a long storage time had little effect on the fluorescence intensity of the NCDs, indicating that the NCDs had good photostability (Figure S3).

3.3 | Optimization of the experimental conditions

3.3.1 | Effect of pH

Figure S2a displays the effect of pH (4.0–11.0) on the fluorescence intensities of the NCDs and NCDs-bilirubin (NCDs-BIL) system, where F_0 and F represent the fluorescence intensities of the NCDs and NCD-BIL system, respectively. The fluorescence intensity of the NCDs shows little change with an increase in the pH in the absence of bilirubin. However, the fluorescence intensity decreases with the

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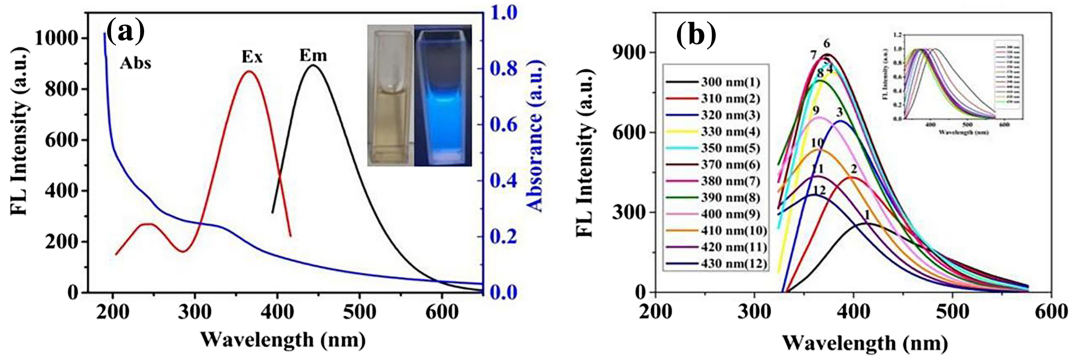

FIGURE 4 (a) UV-visible absorption and fluorescent excitation and emission spectra of the NCDs; (b) fluorescence emission spectra at different excitation wavelengths

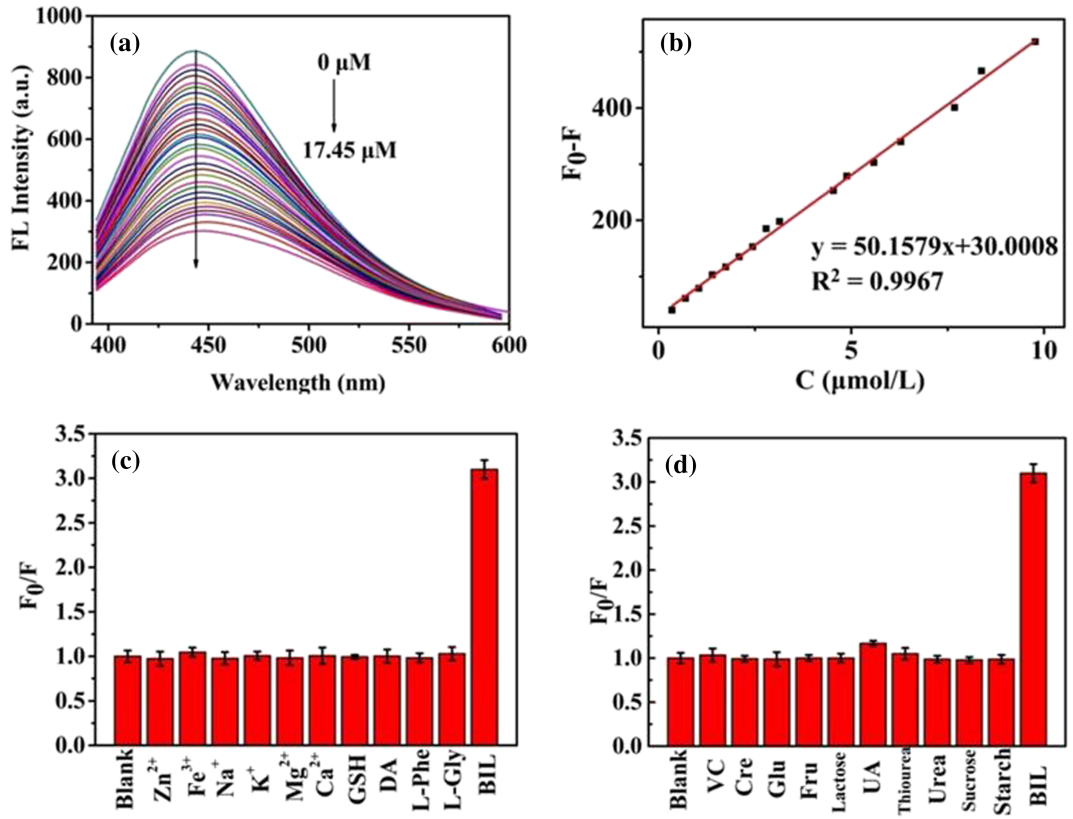

FIGURE 5 (a) The fluorescence quenching of the NCDs at various concentrations of bilirubin; (b) Stern–Volmer linear plot of F_0/F versus the concentration of bilirubin; (c, d) effect of common metal ions and biomolecules on the NCDs

addition of bilirubin. The value of F_0/F increases to maxima at pH 7.8, illustrating the high sensitivity. Thus, a PBS buffer solution with pH 7.8 was selected for use in the subsequent experiments.

3.3.2 | Effect of the NCDs concentration

Furthermore, the effect of NCDs concentration (0.2–0.6 mg/mL) on the NCDs-BIL system was studied using the Stern–Volmer curves as shown in Figure S2b. The slopes of the Stern–Volmer curves increase until the concentration reaches 0.4 mg/mL, and then decrease with further increase in NCDs concentration. The possible reason is that high concentration leads to self-absorption which decreases the fluorescence intensity of NCDs. Thus, 0.4 mg/mL was chosen as the optimal concentration of NCDs for the subsequent experiments.

3.3.3 | Reaction time

The optimal reaction times for the detection of bilirubin were investigated. The fluorescence intensity was monitored by adding a known concentration of bilirubin to the NCDs solution. As shown in

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Figure S2c, the fluorescence intensity initially decreased and was almost unchanged after 3 min. Thus, the optimal reaction time for the detection of bilirubin was chosen as 3 min.

3.4 | Detection of bilirubin and selectivity of the method

With the optimal conditions in hand, the detection of different concentrations of bilirubin was investigated and the results are shown in Figure 5a. We observed that the fluorescence intensity of the NCDs decreased continuously with increased bilirubin concentrations. A good linear response for the NCDs over a concentration range of bilirubin of 0.35 to 9.78 μM was obtained, following the Stern–Volmer behavior. The correlation coefficient R^2 was 0.9967 with a detection limit of 33 nM (Figure 5b). Compared with other detection methods,[29,43–48] the NCDs exhibited an excellent linear range and the lowest detection limit (Table 1). Hence, the NCDs can be used as an effective fluorescence probe for the detection of bilirubin.

To evaluate the selectivity of bilirubin, several common metal cations (Zn^{2+}, Fe^{3+}, Na^{+}, K^{+}, Mg^{2+}, and Ca^{2+}) and different biomolecules (glutathione, dopamine, L-phenylalanine, L-glycine, vitamin C, creatine, glucose, fructose, lactose, uric acid, thiourea, urea, sucrose, and starch) were investigated. Figure 5c,d show that only bilirubin had an

TABLE 1 Comparison of various methods for bilirubin detection

Detection method	Linear range	Detection limit (nM)	Reference
Spectrophotometry	10–125 μM	Not specified	43
Colorimetry	0.25–2 μM	200 ± 19 nM	44
Electrochemistry	2–70 μM	840 nM	45
Electrochemistry	0.2–7 μM	86.32 nM	46
Fluorimetry	0.1–5 μM	70 nM	47
Fluorimetry	25–50 μM	150 nM	48
Fluorimetry	1–50 μM	248 ± 12 nM	44
Fluorimetry	1–10 μM	2030 nM	29
Fluorimetry	0.35–9.78 μM	33 nM	This work

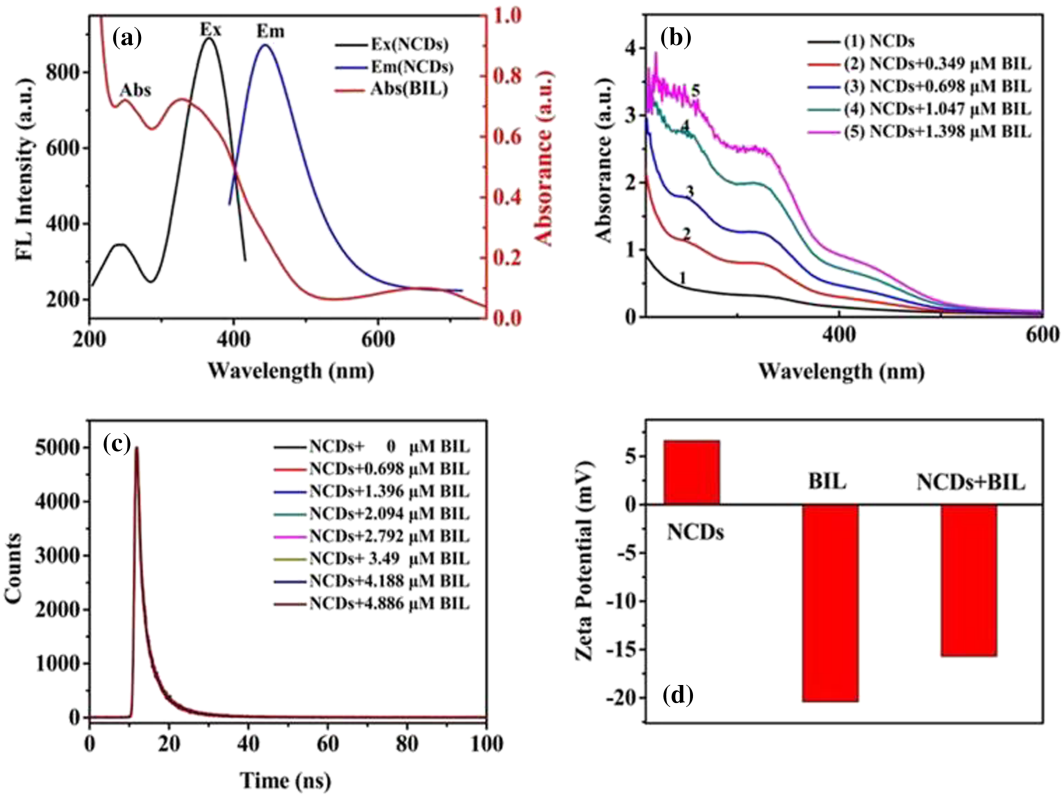

FIGURE 6 (a) The UV-visible absorption, fluorescence excitation, and emission spectra of the NCDs; (b) UV-visible absorption spectra of the NCD solution after addition of various concentrations of bilirubin; (c) fluorescence lifetime in the presence of NCDs and NCDs-BIL system; (d) zeta potentials of the NCDs, bilirubin, and NCDs-BIL system

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appreciable 'turn-off' effect of the fluorescence intensity, indicating that the NCDs had good selectivity for bilirubin. In addition, Figure S3 indicates that the blue fluorescence of the NCDs was almost unchanged by the addition of the earlier ions and biomolecules, under UV light.

3.5 | The mechanism of the fluorescence quenching of bilirubin

To investigate the quenching mechanism, we used UV-visible absorption, fluorescence lifetime, and zeta potential measurements to distinguish static from dynamic quenching. Figure 6a shows the UV-visible absorption, fluorescence excitation, and emission spectra of the NCDs, which have an overlap between them. Figure 6b shows that the UV-visible absorption intensity of the NCDs was enhanced with increased bilirubin concentrations, but the absorption wavelength (330 nm) did not shift. This result indicated that the fluorescence quenching of the NCDs was probably caused by the strong interaction between the NCDs and bilirubin. In addition, the color of the NCDs solution changed from light yellow to dark yellow with an increase in the concentration of bilirubin resulting in a decrease in the light transmission of the NCDs and fluorescence quenching (Figure S4). These results tend to support a static quenching mechanism.

A static quenching mechanism was also indicated by the fluorescence lifetime. If static quenching occurs, the fluorescence lifetime should remain unchanged within the margin of error.[49] As shown in Figure 6c and Table S1, in the absence of bilirubin, the average fluorescence lifetime of the NCDs was 2.99 ns. The average fluorescence lifetime changed little, ranging from 2.87 to 2.96 ns, with the addition of different concentrations of bilirubin, indicating that there was no significant fluorescence resonance energy transfer in the NCDs-BIL system. In other words, the fluorescence change of the NCDs was attributed to electrostatic and hydrogen bonding interactions between the NCDs and bilirubin, while the fluorescence quenching was because of the color change. In addition, the zeta potentials for the NCDs, bilirubin, and the NCDs-BIL system were 6.62, −20.4, and −15.7 mV, respectively (Figure 6d), which further indicated that there was a strong electrostatic interaction between the NCDs and bilirubin. Thus, the quenching mechanism of the NCDs-BIL system was static quenching, and the schematic diagram is shown in Figure S5. NCDs prepared by the hydrothermal process using sucrose and L-proline as raw materials were excited with blue fluorescence at 370 nm. After adding bilirubin, it may react with phenolic –OH, −COOH, and –NH groups present on the surface of NCDs. These complexations facilitate non-radiative electron–hole recombination leading to fluorescence quenching.[28]

3.6 | Determination of bilirubin in biological samples

To expand the applicability, the NCDs were used as a fluorescence probe for biological samples. Different concentrations of bilirubin were added to diluted human urine and blood serum samples, and the content of bilirubin in the samples was determined by the standard addition recovery method. As shown in Table 2, the recovery rates of bilirubin were 95.34–104.66%, and the relative standard deviations (RSDs) were 1.58–4.22% (< 5%), which implied that the synthetic NCDs could be a potential candidate for the determination of low levels of bilirubin in human urine and serum samples.

3.7 | Cytotoxicity and cell imaging of the NCDs as a probe

The cytotoxicity of different concentrations of NCDs toward HeLa cells was determined by the MTT assay. As shown in Figure 7, at NCDs concentrations up to 900 μg/mL, the cell viability was still

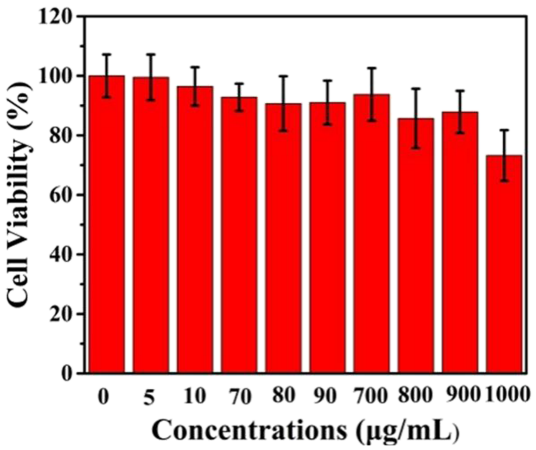

FIGURE 7 Cell viability of HeLa cells in the presence of different concentrations of the NCDs

TABLE 2 Determination of bilirubin in urine and serum samples

Sample	Spiked (μmol/L)	Measured (μmol/L)	Recovery (%)	Relative standard deviation (%) ($n = 3$)
	2.79	2.92	104.66	1.87
Urine	5.58	5.42	96.99	2.81
	8.38	8.44	100.72	1.76
	2.79	2.66	95.34	1.58
Serum	5.58	5.66	101.43	2.71
	8.38	8.54	101.91	4.22

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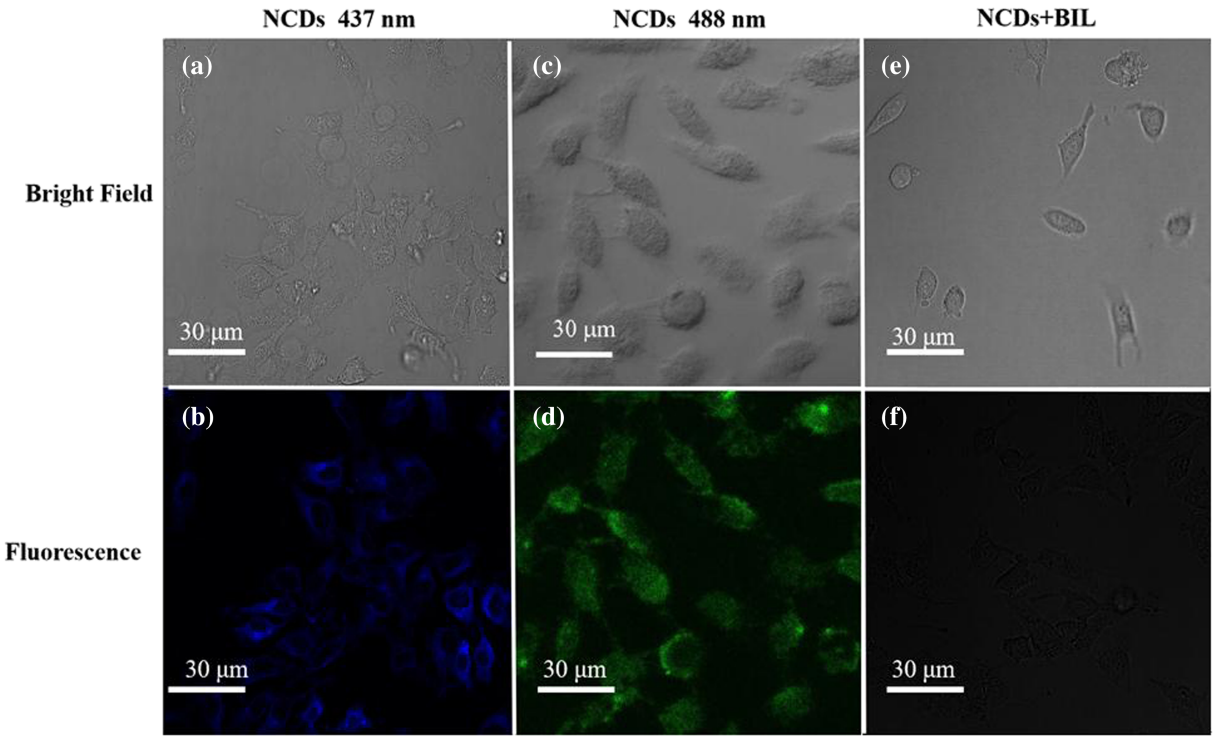

FIGURE 8 The confocal fluorescence images of HeLa cells incubated with NCDs in bright field (a, c) blue channel (b) and green channel (d); and confocal fluorescence images of HeLa cells incubated with NCDs and bilirubin in bright field (e) and blue channels (f)

higher than 80% after incubation for 24 h, indicating the low toxicity of the NCDs. Encouraged by the biocompatibility and low toxicity of the NCDs, we investigated the ability of the NCDs to detect bilirubin in HeLa cells by fluorescence imaging. Figure 8 shows the confocal fluorescent images of HeLa cells incubated with the NCDs or a mixture of NCDs-BIL in bright field (Figure 8a,c,e), blue channel (Figure 8b,f), and green channel (Figure 8d). We observed that the cell morphology was normal. After incubation with NCDs, the HeLa cells showed strong blue (Figure 8b) and green (Figure 8d) fluorescence, indicating that the NCDs had good biocompatibility and low toxicity. When HeLa cells were incubated with NCDs (0.4 mg/mL) and bilirubin (20 μM) at 37°C for 1.5 h, almost no fluorescence was observed in the blue channel (Figure 8f), indicating that bilirubin could completely quench the fluorescence of the NCDs solution. Thus, the NCDs can be used as a 'turn-off' fluorescence probe for the imaging of bilirubin in cells.

4 | CONCLUSIONS

NCDs with bright blue fluorescence were synthesized by a hydrothermal method using sucrose and L-proline as raw materials and characterized by different techniques. The NCDs showed good water solubility, high dispersibility, and optical properties with a fluorescence quantum yield of 23.4%. The NCDs could be used for the detection of bilirubin with a detection limit of 33 nM. The quenching mechanism of the NCD-BIL system was via static quenching. The NCDs were used for bilirubin detection in human urine and serum samples with a recovery of 95.34–104.66%. The low cytotoxicity and good biocompatibility of the NCDs were indicated by an MTT assay and cell imaging of HeLa cells. Compared with other detection systems, using NCDs for bilirubin detection is a facile and efficient method with good sensitivity and selectivity. Thus, this system is expected to have further biological and environmental applications.

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